JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO Sri Annai Agencies

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 36.93 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.673188, 78.001259)

Welcome to the Journey Risk Management Study

To conduct a thorough analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India, to M2F2+4M8, Salem, Tamil Nadu 636013, India, here are the points:

1. **Overview of the Route Map:**
   - The route is approximately 36.93 kilometers long and typically used to connect Sangagiri and Salem. The main roads involved likely include NH 544/45, which are significant highways in the

region.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**

    - Tamil Nadu generally experiences tropical weather, with the southwest monsoon (June to September) and northeast monsoon (October to December). During these periods, heavy rains can lead to flooding and reduced visibility, creating hazardous driving conditions. Summer months can be intensely hot, potentially affecting vehicle performance.

3. **Analysis of Traffic Patterns:**

    - Traffic tends to be heavier during weekday mornings (7:00-10:00 AM) and evenings (5:00-8:00 PM). NH 544 is a major national highway and can experience congestion, particularly near urban centers like Salem.

4. **Assessment of Road Quality and Infrastructure:**

    - The main highways are generally well-maintained, though certain sections may have potholes, especially after heavy rains. Rural and peripheral roads could be narrower with less lighting at night.

5. **Suggestions for Alternative Routes for Emergencies:**

    - An alternative route could involve taking minor roads that run parallel to NH 544, though these may not be suitable for heavy vehicles due to size and weight restrictions. Local roadmaps will provide precise alternatives based on the latest developments.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

    - Regulations in India require permits for the transport of hazardous materials. Compliance with weight restrictions and adherence to designated routes is mandatory. Transport is usually restricted during peak city hours to minimize risk.

7. **Overview of Historical Incidents:**

    - While specific historical data for this route is not readily available, national statistics indicate that road accidents involving heavy vehicles do occur frequently due to overloading and poor road conditions, especially during monsoon seasons.

8. **Environmental Considerations and Sensitive Areas:**

    - The route traverses through areas that may include natural reserves or agricultural land, necessitating caution to prevent spills or contamination. Noise and air pollution regulations must be considered when near populated areas.

9. **Analysis of Communication Coverage:**

    - Major highways like NH 544 generally have good mobile coverage. However, minor road sections and rural areas could potentially have signal dead zones. Verification with local telecom providers is recommended.

10. **Estimated Emergency Response Times:**

    - Proximity to Salem, a larger urban center, suggests relatively quicker emergency response times (generally within 30 minutes). Rural areas may face longer response times (up to an hour or more).

11. **Overall Summary of Risk Assessment:**

- The route is generally structured to support heavy vehicle transport. However, concerns about weather impact, road quality in certain sections, and peak traffic congestion need addressing. Compliance with local regulations is critical for safe and legal transportation of hazardous materials. Regular communication checks and emergency planning are necessary to mitigate potential risks efficiently.

This risk assessment provides a comprehensive guide for truck drivers transporting hazardous materials along this route, with particular emphasis on preparation for weather changes, traffic disruptions, and adherence to regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48366, 77.88781	30 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	High	11.55839, 78.01288	15 KM/Hr
10	Turn	Medium	11.60526, 78.00451	30 KM/Hr
11	Turn	Medium	11.61696, 78.00827	30 KM/Hr
12	Turn	Medium	11.61753, 78.00777	30 KM/Hr
13	Turn	High	11.62660, 78.00986	15 KM/Hr
14	Turn	High	11.62692, 78.00883	15 KM/Hr
15	Turn	Medium	11.64840, 78.00778	30 KM/Hr
16	Turn	Medium	11.64874, 78.00717	30 KM/Hr
17	Blind Spot	Blind Spot	11.67303, 78.00187	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium
13	hospital	Subam Hospital	11.6069532, 78.0042593	30 km/h	Medium
14	hospital	SR Hospital	11.6075985, 78.0070051	30 km/h	Medium
15	hospital	Nesam Medical Centre and Hospital	11.6078647, 78.0073456	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48366, 77.88781

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60526, 78.00451

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62660, 78.00986

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62692, 78.00883

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

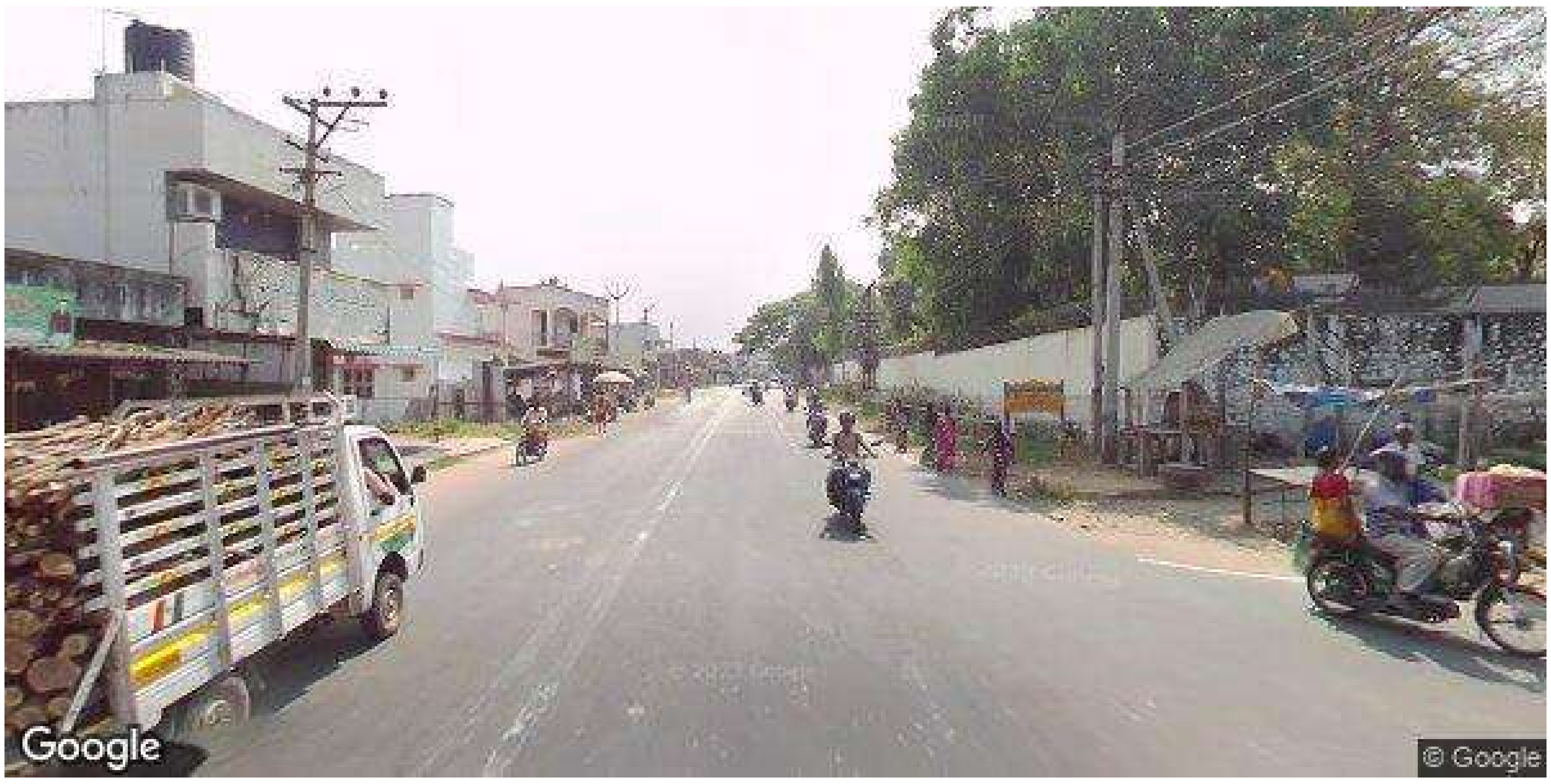
Coordinates: 11.64840, 78.00778

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64874, 78.00717



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67303, 78.00187